

Predictive Energy-Aware Dispatching

Autonomous trams + regenerative braking + section-level flywheel storage

RESEARCH QUESTION

Can real-time dispatch decisions reduce grid energy and peak power while preserving headways and passenger service?

CORE CONCEPT

- A predictive dispatcher forecasts which trams will accelerate or brake in each 750 V DC electrical section.
- It coordinates terminal departures, short holding, eco-coasting, acceleration limits and flywheel charge/discharge targets.
- The controller deliberately creates useful overlap: braking energy from one tram supplies another tram accelerating nearby; surplus energy is captured by the flywheel.

DECISION VARIABLES

- Terminal departure offset (seconds)
- Short stop holding (bounded)
- Eco-coast and acceleration-power limit
- Flywheel SOC target and reserve

COMPETING OBJECTIVES

- Grid energy and rejected regeneration
- Substation peak power
- Headway deviation and bunching
- Passenger waiting and travel time

EXPERIMENTAL DESIGN

- Compare: no storage; reactive SOC control; energy-aware holding; integrated predictive dispatch.
- Run peak/off-peak demand, traffic-signal delay, blocked-track and delayed-departure scenarios on İzmir and Nizhny Novgorod networks.
- Use repeated simulations and Pareto analysis to expose the energy-service trade-off rather than reporting a single tuned case.

KEY EVALUATION METRICS

Operational quality	Energy performance	System robustness
Headway CV · waiting time	Grid kWh · regen reuse	Peak kW · rejected kWh
On-time terminal departures	Flywheel SOC/cycles	Performance under disruptions

SCIENTIFIC NOVELTY

Real-time integration of passenger-oriented headway control, autonomous tram motion and section-level regenerative-energy/flywheel dispatch in a disturbance-prone street network.

MEETING PITCH

- The literature often optimizes metro timetables, speed profiles or storage separately. This paper studies their closed-loop integration for street trams operating by headway under road-traffic disturbances.

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Energy–Service Trade-offs in Tram Control

How much energy can be saved before service quality begins to suffer?

RESEARCH QUESTION

Which headway-regulation strategy gives the best balance between regular arrivals, passenger waiting time and traction energy?

CORE CONCEPT

- Street trams are commonly managed by service intervals rather than a strict arrival time at every intermediate stop.
- Holding and slower driving can restore regularity, but excessive intervention increases travel time and may create queues near terminals.
- The paper quantifies the Pareto frontier between passenger service quality and energy consumption.

CONTROLLERS TO COMPARE

- No active regulation (baseline)
- Distance-headway regulation
- Time-headway regulation
- Terminal-only holding
- Predictive combined regulation

DISTURBANCES TO INJECT

- Variable passenger demand/dwell
- Road-signal delay
- Temporary obstacle or speed limit
- Late terminal departure
- One tram temporarily unavailable

MODEL AND METHOD

- Use the existing passenger, vehicle-mass, curve/switch speed, terminal and traffic-signal models.
- Evaluate identical randomized disturbance seeds for every controller and operating period.
- Report confidence intervals and multi-objective rankings, not only average values.

KEY EVALUATION METRICS

Operational quality	Energy performance	System robustness
Mean/excess waiting	kWh and kWh/passenger-km	Recovery time
Headway CV · bunching rate	Traction/regen balance	Terminal queue length

SCIENTIFIC NOVELTY

A passenger-aware comparison of spatial, temporal and terminal-based headway control with realistic tram physics and energy accounting—not a fixed metro timetable study.

EXPECTED PUBLISHABLE RESULT

- Identify a practical ‘knee point’: a controller that captures most energy savings and regularity benefits without materially increasing passenger waiting.
- Produce reusable benchmark disturbances and clearly defined metrics for later energy-storage and HIL papers.

MEETING PITCH

- This is the fastest paper to execute because the simulator already contains the required operational components; the main work is rigorous experiment design, calibration and statistical comparison.

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Hardware-in-the-Loop Digital Twin Testbed

One control architecture for a physical Arduino tram and a city-scale virtual network

RESEARCH QUESTION

Can a low-cost physical prototype and its digital twin share control logic while remaining safe under sensor, communication and infrastructure faults?

CORE CONCEPT

- The physical tram supplies RFID detections, obstacle distance, motor/door state and radio messages.
- The digital twin supplies the remaining network, passengers, signals, switches, energy sections and dispatcher commands.
- The same C/C++ control core is compiled for embedded hardware, WebAssembly visualization and headless experiments.

PHYSICAL ↔ DIGITAL LOOP

- RFID → synchronized position
- Sensor distance → obstacle state
- PWM/brake command → motion model
- Switch request → infrastructure state
- Telemetry → live twin dashboard

FAULT-INJECTION CAMPAIGN

- Packet delay, loss and corruption
- Missed or duplicated RFID read
- Stuck switch / conflicting authority
- False-positive obstacle detection
- Dispatcher or radio-link loss

VALIDATION PLAN

- Calibrate simulated acceleration, braking distance and stop accuracy against repeated physical runs.
- Measure command-to-response latency, localization error, fail-safe stopping time and recovery behaviour.
- Verify safety invariants: no conflicting switch movement, no unsafe block entry, doors only at platforms and safe fallback after communication loss.

KEY EVALUATION METRICS

Operational quality	Energy performance	System robustness
Stop/position error	Physical vs simulated energy	Fault detection latency
Run-time/headway agreement	PWM/current correlation	Safe-stop success rate

SCIENTIFIC NOVELTY

An accessible distributed control-in-the-loop platform that connects real embedded tram hardware to network-scale passenger, infrastructure and energy operation—not only a vehicle or power-electronics HIL bench.

RESEARCH VALUE AND DEMONSTRATOR

- Produces both a formal validation paper and a visually strong live demonstration for laboratories, competitions and industry meetings.
- Provides the experimental bridge needed to justify the term ‘digital twin’ rather than simulation alone.

MEETING PITCH

- The platform is intentionally low-cost and reproducible, allowing research groups to test autonomous tram control, infrastructure cooperation and faults without access to a full-size vehicle.

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